

***B.Tech. Degree I Semester Regular and Supplementary/
II Semester Supplementary Examination November 2017***

**CE/EE/ME/SE GE 15-1104 A AND CS/EC/IT GE 15-1204 B
BASIC CIVIL ENGINEERING
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. Write brief notes on the following.
- (a) Properties of OPC.
 - (b) Grading of aggregates.
 - (c) Placing of concrete.
 - (d) Isolated footing.
 - (e) Queen closer and ½ Bat.
 - (f) First principle of survey.
 - (g) Foresight.
 - (h) Reduced level.
 - (i) Calculate the F.A.R of a 2 storied building situated in a plot of 20 m × 30 m size having two floors. Ground floor area 100 m² and first floor area 75 m².
 - (j) What is the basic concept of intelligent buildings?

PART B

(4 × 10 = 40)

- II. Describe the different types of cement used in construction.
- OR**
- III. What are the different tests available to check the quality of bricks?
- IV. Draw a detailed sketch of a King post truss used for a span of 8 meters.
- OR**
- V. Draw to a suitable scale, the plan for two layers of 1½ brick thick English bond for a corner of building. Also show the elevation.
- VI. Following consecutive readings were taken with a dumpy level and a 4 meter leveling staff
0.600, 2.100, 1.850, 3.800, 2.000, 1.635, 3.125, 3.750, 0.750, 1.500, 2.000
The level was shifted after the 4th and the 8th readings. The first reading was taken on a B.M of R.L 100. Calculate the reduced levels of all the points. Also apply the usual check.
- OR**
- VII. Explain the different instruments used in chain survey with neat sketches.
- VIII. Assume you are one of the leading civil engineering consultants in your locality. A party wants to purchase some land for the construction of his residence in that area. If he approaches you, advice him with relevant building rules.
- OR**
- IX. Draw the cross section of a road and explain the classification of roads.

